

SAM, Kazakhstan

WMO: 359250

Lat: 45.024N

Lon: 55.748E

Elev: 289

StdP: 14.54

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.1	-6.0	-17.0	2.2	-11.3	-11.3	3.0	-5.0	25.7	15.7	22.4	21.0	5.1	90	0.563

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.1	102.9	66.7	99.9	65.9	96.9	65.3	70.7	92.0	68.8	90.1	67.4	89.2	7.9	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.1	90.5	77.0	61.7	83.2	77.0	59.4	76.3	76.4	34.7	92.0	33.1	89.3	31.9	88.8	83.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 22.8	20.9	18.9														
(5)			DB	-15.8	107.5	9.5	3.6	-22.7	110.1	-28.2	112.3	-33.6	114.3	-40.5	116.9	
			WB	-15.8	74.3	9.3	3.5	-22.5	76.8	-28.0	78.9	-33.2	80.8	-40.0	83.3	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	51.1	18.0	20.9	36.8	54.3	67.8	77.9	82.2	79.5	65.6	50.3	34.4	23.4		
	DBStd	24.62	11.76	12.52	10.54	9.37	7.82	6.91	6.17	7.08	7.92	8.80	9.88	10.34		
	HDD50	3692	992	814	421	60	1	0	0	0	2	106	470	826		
	HDD65	6724	1457	1234	875	339	61	3	0	2	86	459	917	1291		
	CDD50	4094	0	0	11	189	553	838	999	916	469	116	3	0		
	CDD65	1651	0	0	0	18	148	392	534	453	103	3	0	0		
	CDH74	22156	0	0	3	251	1789	5234	7452	6075	1279	73	0	0		
	CDH80	12032	0	0	0	71	733	2887	4353	3485	494	9	0	0		
(14) Wind	WSAvg	7.8	8.2	8.5	9.6	9.2	8.1	7.5	7.0	6.8	6.6	7.1	7.4	8.0		
(15) Precipitation	PrecAvg	5.6	0.4	0.4	0.5	0.5	0.8	0.7	0.5	0.1	0.2	0.4	0.6	0.4		
	PrecMax	7.6	0.9	1.2	1.5	1.5	1.6	2.1	1.4	0.5	1.2	1.2	1.3	0.9		
	PrecMin	3.2	0.0	0.0	0.0	0.1	0.2	0.1	0.0	0.0	0.0	0.0	0.1	0.0		
	PrecStd	1.2	0.2	0.3	0.4	0.4	0.4	0.6	0.4	0.2	0.3	0.4	0.4	0.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.2	47.8	68.6	87.5	96.2	104.8	107.0	106.3	97.8	80.7	61.5	46.6		
		MCWB	37.0	41.6	52.5	60.7	62.5	66.0	67.9	68.0	64.8	56.8	50.4	39.8		
	2%	DB	36.7	42.6	61.4	81.2	91.7	100.2	103.1	102.3	89.7	75.4	56.3	41.3		
		MCWB	34.9	38.9	49.4	58.8	61.5	65.2	67.1	66.9	60.8	55.6	48.2	37.9		
	5%	DB	34.3	38.3	56.7	75.7	87.6	97.0	100.2	98.9	85.3	70.9	52.2	37.6		
		MCWB	32.8	35.0	46.7	56.8	60.5	64.6	66.5	65.7	59.6	53.1	45.8	35.1		
	10%	DB	32.6	35.1	51.6	70.9	83.7	93.8	96.9	95.5	81.3	66.0	47.7	34.9		
		MCWB	31.2	32.6	43.8	54.4	59.8	63.7	65.7	65.1	58.1	51.3	42.9	33.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	37.9	43.2	54.0	65.8	67.7	71.3	73.4	73.4	67.2	60.3	54.4	41.8		
		MCDB	39.6	46.3	66.3	78.8	84.6	92.6	93.7	97.0	88.0	71.9	58.4	45.1		
	2%	WB	35.0	38.9	50.3	61.4	64.8	68.6	70.8	70.2	63.3	57.3	49.6	38.4		
		MCDB	36.4	42.4	59.5	75.6	81.7	89.1	91.8	92.7	84.3	70.5	54.4	40.7		
	5%	WB	33.1	35.4	47.3	58.0	62.9	66.9	69.2	68.0	61.1	55.1	46.2	35.5		
		MCDB	34.2	37.8	55.7	72.0	81.3	88.3	90.2	90.4	80.8	68.1	51.1	37.1		
	10%	WB	31.3	32.8	44.2	55.3	61.3	65.5	67.8	66.4	59.5	52.6	43.2	33.1		
		MCDB	32.5	34.8	51.1	69.0	79.4	87.3	89.5	90.4	77.9	64.4	47.8	34.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	12.8	15.1	17.1	21.5	22.8	23.8	23.1	24.4	24.6	22.5	15.5	12.2		
		MCDBR	9.8	15.8	24.9	27.9	27.8	28.5	27.0	28.7	30.1	29.2	22.1	14.9		
	5% WB	MCWBR	8.6	12.0	14.5	12.7	10.0	9.2	9.0	10.2	11.9	14.1	14.4	11.8		
		MCWBR	9.3	13.9	23.6	25.0	23.7	23.9	22.6	24.7	25.8	24.9	19.1	12.6		
(40) Clear-Sky Solar Irradiance	taub	taub	0.361	0.384	0.457	0.509	0.502	0.484	0.469	0.453	0.419	0.416	0.382	0.371		
		taud	2.044	2.014	1.914	1.823	1.891	1.981	2.042	2.064	2.104	2.050	2.099	2.046		
	Ebn at Noon	Ebn at Noon	221	242	240	239	246	252	254	253	250	227	212	201		
		Edh at Noon	33	42	53	63	61	56	52	49	43	40	31	30		
(44) All-Sky Solar Radiation	RadAvg	382	687	1125	1615	2021	2275	2181	1976	1529	984	550	334			
	RadStd	73	121	112	145	139	137	90	107	88	63	69	60			

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	+2.16	-1.53	-2.05	N/A	N/A	N/A	N/A
(47)	Regional (1 neighbor)	N/A	N/A	N/A	+2.16	-1.53	-2.05	N/A	N/A	N/A	N/A

Nomenclature: See separate page